

Yohan Vergis Vinu

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EDUCATION

University of Pennsylvania

Master of Science in Engineering, Computer and Information Science (GPA: 3.95/4)

Philadelphia, PA

Aug. 2025 – May 2027

SRM Institute of Science and Technology

Bachelor of Technology, Computer Science and Engineering (Top 1%, GPA: 9.80/10)

Chennai, India

Aug. 2021 – May 2025

EXPERIENCE

Research Assistant, Penn NLP Group

University of Pennsylvania (Advised by C. Callison-Burch, M. Yatskar)

Jun. 2026 – Present

Philadelphia, PA

- Building intervention harnesses that keep models acting on **safety-critical facts** disclosed earlier in a conversation, raising task success from **48% to 87%** on a safety benchmark while cutting false alarms below baseline (**11% to 8%**)
- Decomposed memory systems (**RAG, GraphRAG, HippoRAG, MemGPT**) into component modules and benchmarked each, showing retrieval-based memory matches no memory when the query carries no retrieval cue

Graduate Teaching Assistant

University of Pennsylvania

Jan. 2026 – Present

Philadelphia, PA

- Supported **500+** students across CIS 5210 (Artificial Intelligence) and CIT 5960 (Algorithms and Computation); co-designed assignments and rubrics with course staff and graded exams

Software Engineer Intern, Applied AI

Moative (early-stage AI startup) | Sole engineer on client projects

Aug. 2024 – Jun. 2025

Chennai, India

- Automated recruiter screening at **15+ companies** across **100+ interviews** by shipping a multi-tenant HR platform (**Django, LangGraph**) running diarization, extraction, and scoring agents in parallel over audio or text input, cutting screening time by **70%**; deployed on ECS with staging/production CI pipelines
- Designed an async **FastAPI** company-enrichment service with batch processing, connection pooling, and exponential backoff retries on scrapes, with agents classifying firms to standardized NAICS codes at 95% accuracy through retrieval
- Shipped a production multi-channel support agent handling outage, billing, and account workflows across **voice and SMS (Flask, Twilio, ElevenLabs, Whisper)**, with checkpointing for cross-session memory
- Developed a permitting agent that validated site plans and engineering drawings against municipal code via a LangGraph document-verification workflow, cutting a 2–3 week review to minutes in demo

Machine Learning Engineer Intern

Samsung Research

Aug. 2023 – Mar. 2024

Chennai, India

- Built a pet facial-state classifier (eye, ear, mouth position) on a self-annotated 10K+ image dataset; fine-tuned MobileNetV2-SSD **beat baseline by 11%**, ahead of YOLOv8 and ResNet; awarded Samsung's Certificate of Excellence

PROJECTS

ModForge: Natural-Language Minecraft Mod Generator | *FastAPI, Next.js, Docker, Java*

2026

- Built a fully autonomous generate-compile-repair loop: an agent writes **Fabric/Java** against retrieved API context, a sandboxed **Docker + Gradle** container compiles under resource limits, and build errors feed back for self-repair
- Reverse-verified Minecraft 26.1's newly unobfuscated API directly from compiled JARs after official mappings were discontinued; shipped the surrounding product with usage-metered auth and one-click GitHub OAuth export

LLMFuzz: Distributed LLM-Guided Fuzzing Agent | *Claude SDK, Redis Streams, coverage.py*

2026

- Outperformed a random-fuzzing baseline by **7.6 pp** (98.3% vs 90.7% branch coverage) with **53% fewer inputs**, surfacing **4 crashes** it missed, via an **LLM** that reads source and generates inputs against live branch-coverage feedback
- Scaled fuzzing across stateless workers by building a Redis Streams consumer group; coordinator auto-reassigns low-coverage targets to a fallback strategy

PennSearch: Distributed Cloud Search Engine | *Java, AWS EC2*

2025

- Built a **search engine** over **300k+** crawled pages with multi-signal ranking (TF-IDF, PageRank, title/URL matching), serving **sub-50ms** queries from an in-memory inverted index
- Implemented the underlying distributed infrastructure from scratch: a custom HTTP server, a consistent-hashing key-value store with cross-node replication, and a Spark-like parallel analytics engine (map/flatMap/foldByKey/join)

TECHNICAL SKILLS

Programming Languages: Python, SQL, Java, TypeScript, JavaScript, C/C++

AI/ML: LangGraph/LangChain, LangSmith, NumPy, Pandas, PyTorch, Hugging Face, TensorFlow, OpenCV

Web Dev. & APIs: Django, FastAPI, Flask, React, Next.js, REST APIs

Cloud, Databases & DevOps: AWS, PostgreSQL, Docker, Git/GitHub, GitHub Actions, Vector DBs, GCP, Redis